

Visualization results

Results for the MiniTorch visualization tasks.

Task 0.5 — Module 0

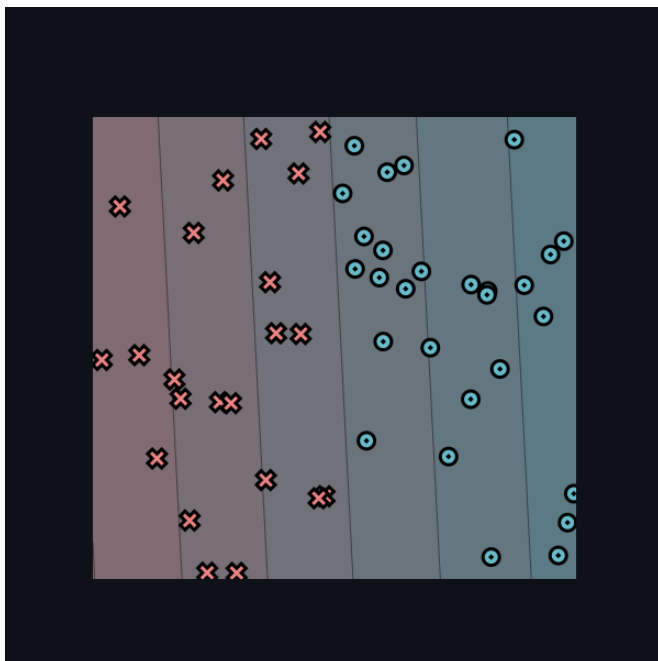
Dataset visualization and hand-created linear classifiers.

Simple

Parameters:

```
weight_0_0 = -1.21
weight_1_0 = -0.06
bias_0 = 0.41
```

Result:



Task 1.5 — Scalar training

Training results for the scalar model.

Dataset	Hidden layers	Learning rate	Epochs	Final loss	Correct
Simple					
Diag					
Split					

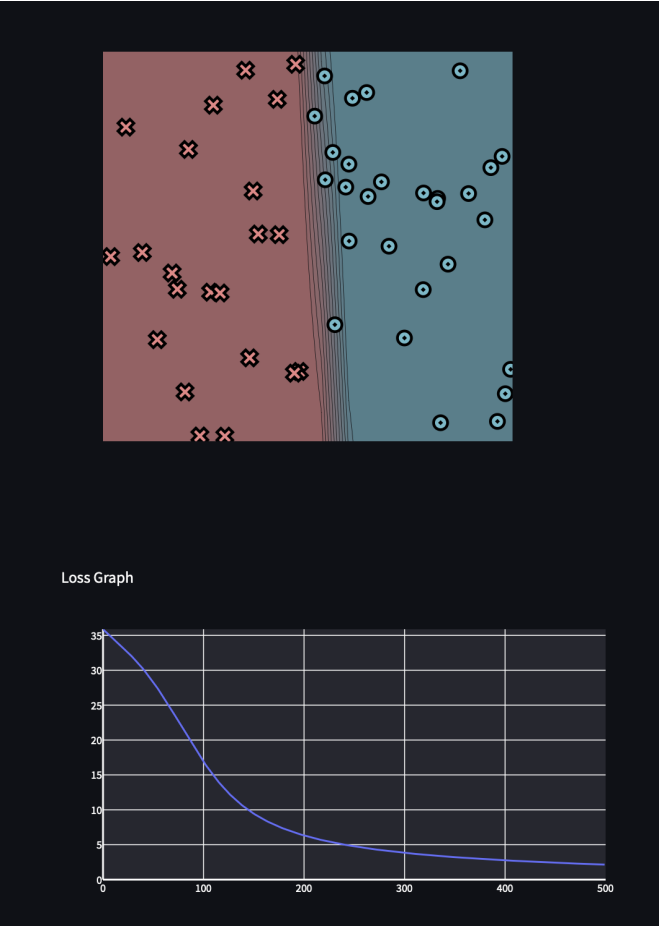
Dataset	Hidden layers	Learning rate	Epochs	Final loss	Correct
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Xor

Screenshots:

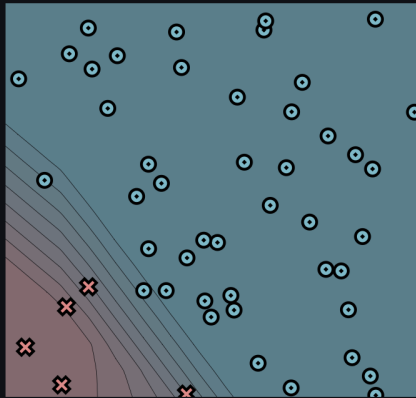
Simple

Epochs	Final loss	Correct
500	1.5319677969	50/50

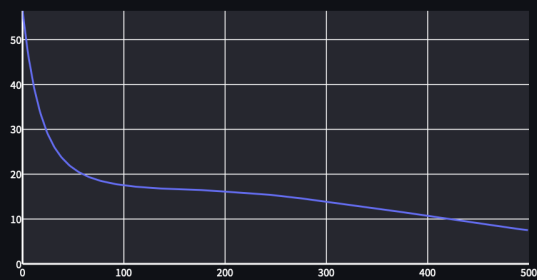


Diag

Epochs	Final loss	Correct
500	30.3233157022	29/50

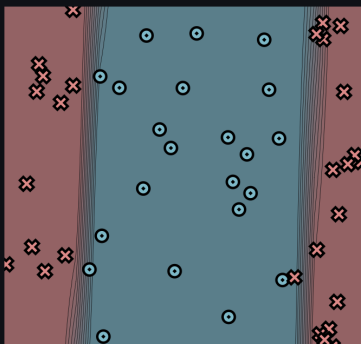


Loss Graph

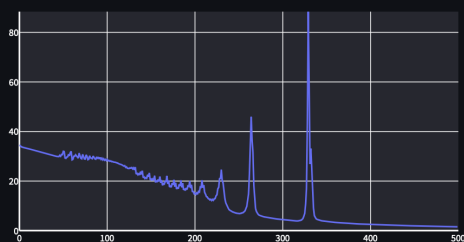


Split

Epochs	Final loss	Correct
500	14.3492259151	42/50

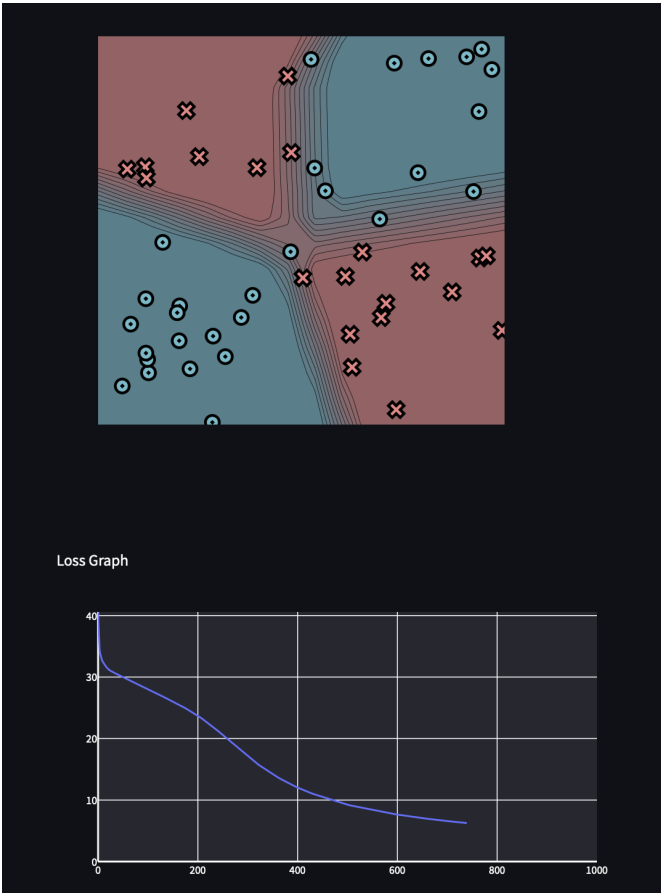


Loss Graph



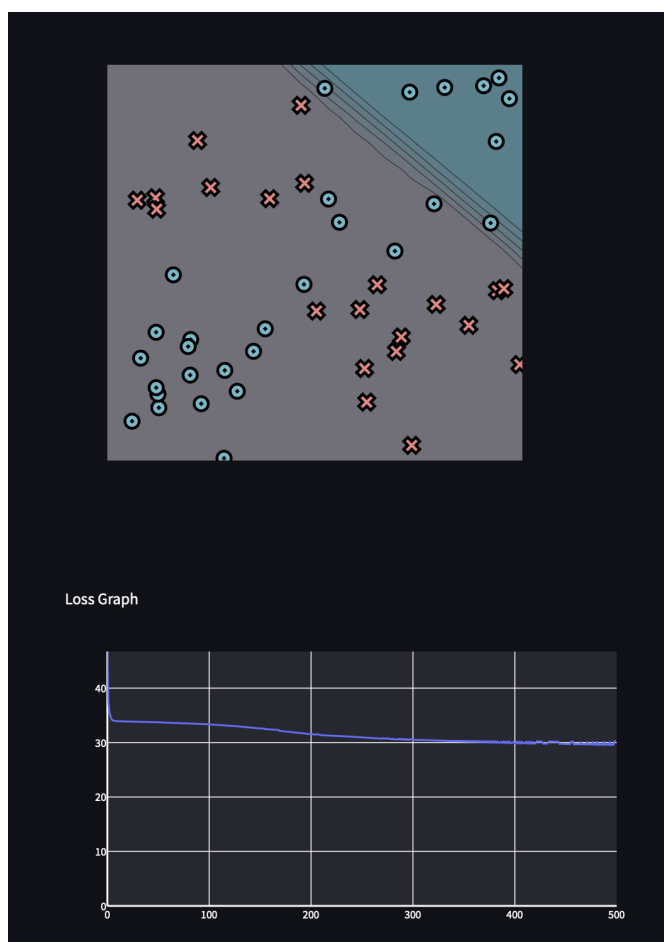
Xor, hidden layer 10

Epochs	Final loss	Correct
500	17.4706381998	44/50



Xor, hidden layer 3

Epochs	Final loss	Correct
500	20.3850323176	37/50

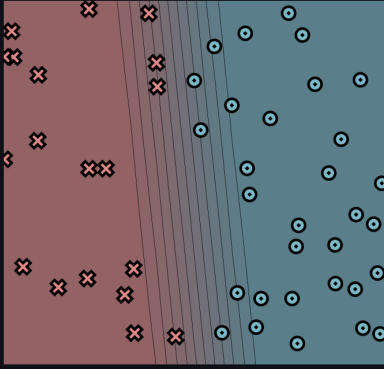


Task 2.5 — Tensor training

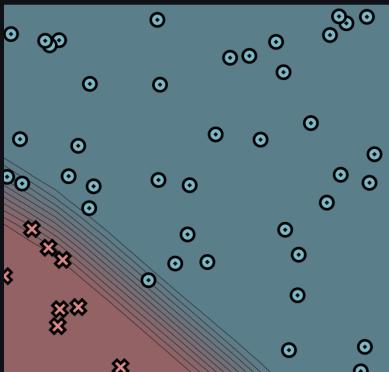
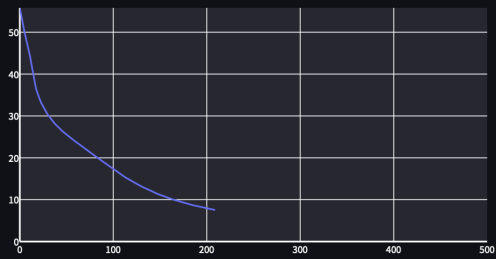
Training results for the tensor model.

Dataset	Hidden layers	Learning rate	Epochs	Time per epoch	Final loss	Correct
Simple	10	0.05	1000	-	4.6856	50/50
Diag	10	0.05	500	—	3.2544	49/49
Split	10	0.1	1000	-	4.2522	48/50
Xor	10	0.05	1000	—	9.4129	47/49

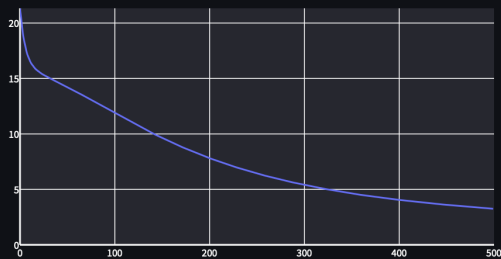
Screenshots:

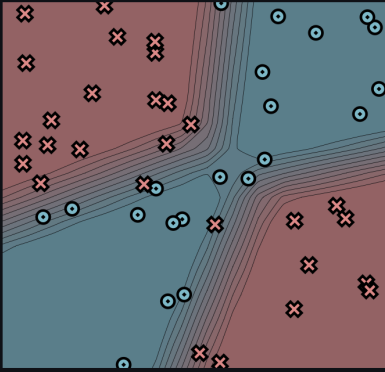


Loss Graph

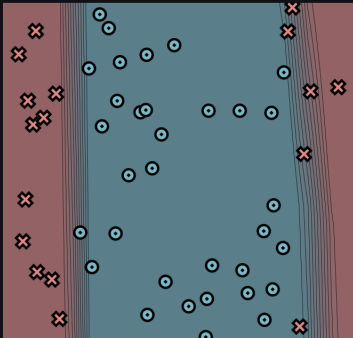
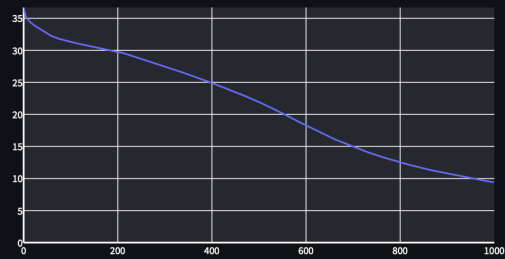


Loss Graph





Loss Graph



Loss Graph

